1 Introductory Graph Theory

1.1 Graphs and Subgraphs

Definition 1.1. *A graph G is a pair $G = (V(G), E(G))$, where $V(G)$ is a set of elements called vertices and $E(G)$ is a set of 2-element subsets of $V(G)$ called edges. If there is no chance of confusion, then we may write V and E instead of $V(G)$ and $E(G)$. Also, an edge $\{u, v\} \in E$ is denoted by uv.*

Definition 1.2. *Let $G = (V, E)$ be a graph and let $uv \in E$. (i) We say that uv joins u and v, that u and v are adjacent, that u is a neighbor of v (and vice versa), and that uv is incident to u (and v). (ii) The neighbourhood of $u \in V$, denoted as $N(u)$, is the set of all vertices adjacent to u. The closed neighbourhood of u, denoted $N[u]$, is the neighbourhood of u together with u itself. That is, $N(u) = \{v \in V : uv \in E\}$ and $N[u] = \{v \in V : uv \in E\} \cup \{u\}$.*

Definition 1.3. *Let G be a graph. The order of G is the number of vertices in G. Conventionally, we use $n = |V(G)|$. The size of G is the number of edges in G. Conventionally, we use $m = |E(G)|$. We say that G is an (n, m) graph if $n = |V(G)|$ and $m = |E(G)|$. A graph G is complete if every pair of vertices in G is adjacent. The complete graph on n vertices is denoted K_n. If a graph G has no edges, then it is empty.*

Definition 1.4. *Let G and H be graphs. H is a subgraph of G, denoted as $H \subseteq G$, if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.*

Definition 1.5. *Let G be a graph and $S \subseteq V(G)$. The subgraph of G induced by S, denoted by $G[S]$, is defined by $V(G[S]) = S$ and $E(G[S]) = \{uv \in E(G) : u \in S, v \in S\}$.*

Definition 1.6. *Let G be a graph and $S \subseteq E(G)$. The subgraph of G induced by S, denoted by $G[S]$, is defined by $V(G[S]) = \{v \in V(G) :$ there exists some edge $uv \in S\}$ and $E(G[S]) = S$.*

Definition 1.7. *Let G be a graph and let $H \subseteq G$. H is a spanning subgraph of G if $V(G) = V(H)$.*

1.2 Degrees and Degree Sequences

Definition 1.8. *Let G be a graph. The degree of a vertex $v \in V$, denoted $\deg_G v$ or just $\deg v$ when there can be no confusion, is the size of the neighbourhood of v. That is, $\deg_G v = |N(v)|$. If every vertex in G has degree d, then G is said to have degree d, denoted as $\deg_G = d$. If $\deg_G = d$, then G is said to be regular or d-regular.*

Definition 1.9. *Let G be a graph. The minimum degree of G is defined as $\delta(G) = \min\{\deg v : v \in V\}$. The maximum degree of G is defined as $\Delta(G) = \max\{\deg v : v \in V\}$.*

Definition 1.10. *Let G be a graph with $V = \{v_1, \ldots, v_n\}$ such that $\deg v_1 \geq \cdots \geq \deg v_n$. Then $\deg v_1, \ldots, \deg v_n$ is called the degree sequence of G.*

1.3 Walks and Components

Definition 1.11. *Let G be a graph. (i) A walk is a sequence of vertices $v_0, \ldots, v_k$ such that $v_i v_{i+1} \in E(G)$ for $i \in \{0, \ldots, k-1\}$, and is referred to as a $v_0 - v_k$ walk. The length of the walk $v_0, \ldots, v_k$ is k. The walk $v_0, \ldots, v_k$ is closed if $v_0 = v_k$, otherwise it is open. (ii) A trail is a walk $v_0, \ldots, v_k$ in which every edge $v_i v_{i+1}$ for $i \in \{0, \ldots, k-1\}$ is distinct. (iii) A circuit is a closed trail. (iv) A path is a trail $v_0, \ldots, v_k$, denoted as $P_{v_0 v_k}$, in which every vertex v_i is distinct for $i \in \{0, \ldots, k-1\}$. (v) A cycle is a closed path.*

Definition 1.12. *A graph G is connected if there exists a path P_{uv} for all $u, v \in V(G)$. If G is not connected, then it is disconnected. Each maximal connected subgraph of G is called a component of G.*

1.4 Compliments and Combinations of Graphs

Definition 1.13. *Let G be a graph. The complement of G, denoted $\overline{G}$, is the graph defined by $V(\overline{G}) = V(G)$ and $E(\overline{G}) = \{uv : u, v \in V(G), uv \notin E(G)\}$.*

Definition 1.14. *Let G and H be graphs. The union of G and H, denoted $G \cup H$, is the graph defined by $V(G \cup H) = V(G) \cup V(H)$ and $E(G \cup H) = E(G) \cup E(H)$.*

Definition 1.15. *Let G be a graph. The graph kG, where k is a positive integer, is defined by $V(kG) = V(G) \times \{1, \ldots, k\}$ and $E(kG) = \{(u, i)(v, i) : uv \in E(G),\ i \in \{1, \ldots, k\}\}$. The scalar multiple kG is simply k disconnected copies of G.*

Definition 1.16. *Let G and H be disjoint graphs. The join of G and H, denoted $G + H$, is the graph defined by $V(G + H) = V(G) \cup V(H)$ and $E(G + H) = E(G) \cup E(H) \cup \{uv : u \in V(G), v \in V(H)\}$.*

Definition 1.17. *Let G and H be graphs. The cartesian product of G and H, denoted $G \times H$, is the graph defined by $V(G \times H) = V(G) \times V(H)$ and $E(G \times H) = \{(g, h)(g', h) : gg' \in E(G)\} \cup \{(g, h)(g, h') : hh' \in E(H)\}$.*

1.5 Mappings between graphs

Definition 1.18. *Let G and H be graphs. Let f be a mapping $f : V(G) \to V(H)$. (i) f is a homomorphism if $f(u)f(v) \in E(H)$ for every $uv \in E(G)$. (ii) f is an isomorphism if f is a bijection, $f(u)f(v) \in E(H)$ for every $uv \in E(G)$, and $f(u)f(v) \notin E(H)$ for every $uv \notin E(G)$. G and H are said to be isomorphic, denoted $G \cong H$. (iii) f is an automorphism if $G = H$ and f is an isomorphism.*

Definition 1.19. *Let G be a graph. G is vertex-transitive if for every $u, v \in V(G)$ there exists an automorphism f of G such that $f(u) = v$.*

1.6 Digraphs

Definition 1.20. *A digraph D is a pair $D = (V(D), A(D))$ where $V(D)$ is a set of elements called vertices, and $A(D)$ is a set of elements from $V(D) \times V(D)$, called arcs.*

Definition 1.21. *Let D be a digraph and let $u \in V(D)$. (i) The indegree of u is the number of vertices from which u is adjacent, denoted $\deg^+(u)$. The outdegree of u is the number of vertices to which u is adjacent, denoted $\deg^-(u)$. (ii) The set of all vertices which are adjacent from u is denoted $N^+(u)$, and the set of all vertices which are adjacent to u is denoted $N^-(u)$. (iii) For a given vertex u, the set of all arcs vu is denoted $A^+(u)$, the set of all arcs uv is denoted $A^-(u)$. A vertex with indegree zero is called a source and a vertex with outdegree zero is called a sink.*

Definition 1.22. *Let D be a digraph. The underlying graph of D is the graph G defined by $V(G) = V(D)$ and $E(G) = \{uv : uv \in A(D)$ or $vu \in A(D)\}$.*

1.7 Cut-sets and Blocks

Definition 1.23. *Let G be a graph. Let $v \in V(G)$ and $e \in E(G)$. v is a cut-vertex if $G - v$ has more components than G. e is a bridge if $G - e$ has more components than G.*

Definition 1.24. *Let G be a graph. G is separable if it contains one or more cut-vertices. G is non-separable if it contains no cut-vertices. A block is a maximal non-separable subgraph.*

1.8 Distance in Graphs

Definition 1.25. *Let G be a connected graph and let $u, v \in V(G)$. The distance from u to v, denoted $\mathrm{d}(u, v)$, is defined as the length of a shortest $u-v$ path in G. If no path exists, we set $d(u, v) = \infty$. The diameter of G is $\max\{d(u, v) : u, v \in V(G)\}$, and the radius of G is $\min\{\max\{d(u, v) : v \in V(G)\} : u \in V(G)\}$.*

2 Connectivity

2.1 Vertex-cuts and edge-cuts

Definition 2.1. *Let G be a graph. (a) Let $S \subseteq V(G)$. S is a vertex-cut of G if $G - S$ contains more components than G. (b) G is k-connected if G has no vertex-cut of size less than k, i.e. $\kappa(G) \geq k$, where $\kappa(G)$ denotes the connectivity of G. (c) Let $S \subseteq E(G)$. S is an edge-cut of G if $G - S$ contains more components than G. (d) G is k-edge-connected if G has no edge-cut of size less than k, i.e. $\lambda(G) \geq k$, where $\lambda(G)$ denotes the edge-connectivity of G.*

2.2 k-connected graphs and k-edge-connected graphs

Definition 2.2. *Let G be a connected graph. For an integer $k \geq 0$: (a) G is k-connected if $\kappa(G) \geq k$. (b) G is k-edge-connected if $\lambda(G) \geq k$.*

2.3 Menger's Theorem

Definition 2.3. *Let u and v be distinct vertices in a connected graph G. A $u-v$ separating set is a vertex-cut $S \subseteq V(G)$ such that u and v are in different components of $G - S$.*

3 Eulerian and Hamiltonian Graphs

3.1 Eulerian graphs

Definition 3.1. *A (multi-)graph G is Eulerian if it contains an Eulerian circuit (a cycle containing all edges of G). If G contains an Eulerian trail (a walk containing all edges of G but not necessarily closed), then G is said to have an Eulerian trail.*

3.2 Hamiltonian graphs

Definition 3.2. *A graph G is Hamiltonian if it contains a Hamiltonian cycle (a cycle containing every vertex of G exactly once). If G contains a Hamiltonian path (a path containing every vertex of G exactly once), then G is said to have a Hamiltonian path.*

4 Planar Graphs

4.1 Jordan's Theorem and Euler's Formula

Definition 4.1. *A planar graph is a graph that can be embedded in the plane without any edges crossing. An embedding of G in the plane is an assignment of each vertex of G to a distinct point in the plane and each edge to a continuous arc connecting its endpoints, such that no two arcs intersect except possibly at common endpoints.*

Definition 4.2. *A face of a plane graph G is a maximal region of the plane bounded by edges of G. The boundary of a face F is the subgraph of G consisting of the edges and vertices that lie on the boundary of F.*

4.2 Maximal planar graphs

Definition 4.3. *A planar graph is maximal planar if no more edges can be added to it without losing planarity. Equivalently, a maximal planar graph (with $n \geq 3$) has exactly $3n - 6$ edges.*

4.3 Chvátal's Art Gallery Theorem and Fáry's Theorem

Definition 4.4. *An art gallery is represented by a polygon (possibly with holes). The art gallery theorem states that $\lfloor n/3 \rfloor$ guards are necessary and sufficient to cover an n-vertex polygon. (This is background context; a formal definition is omitted.)*

Definition 4.5. *Fáry's Theorem: Every planar graph can be drawn with straight-line edges. (Again, stated for context; no formal definition needed.)*

5 Graph Colourings

5.1 Introduction to vertex colourings

Definition 5.1. *A proper vertex-colouring of a graph G is an assignment of colors to each vertex such that no two adjacent vertices share the same color. The chromatic number $\chi(G)$ is the minimum number of colors needed in a proper coloring of G.*

Example 5.2. *Figure 5.1 shows two proper colourings of the same graph G using the colours white, black, light grey, and dark grey: On the left, a proper 4-colouring; on the right, a proper 3-colouring.*

Example 5.3. *In Figure 5.2, two labellings of the graph C_6 are shown. The one on the left produces a proper 2-colouring, while the one on the right produces a proper 3-colouring. This illustrates that the performance of the greedy algorithm depends on the labeling of vertices chosen.*

5.2 Critically and minimally k-chromatic graphs

Definition 5.4. *A graph G is k-critical if $\chi(G) = k$ and every proper subgraph of G has chromatic number less than k. A graph is minimally k-chromatic if $\chi(G) = k$ and removing any edge or vertex reduces the chromatic number.*

5.3 Brooks' Theorem

Definition 5.5. *Brooks' Theorem: In a connected graph G that is not an odd cycle or a complete graph, $\chi(G) \leq \Delta(G)$, where $\Delta(G)$ is the maximum degree. (Stated for context.)*

5.4 Further results on vertex-colourings

Definition 5.6. *A clique in a graph G is a set of mutually adjacent vertices. The clique number $\omega(G)$ is the size of the largest clique in G. We have $\chi(G) \geq \omega(G)$ for any graph G.*

5.5 Introduction to edge-colourings

Definition 5.7. *A proper edge-colouring of a graph G is an assignment of colors to each edge such that no two adjacent edges share the same color. The edge-chromatic number $\chi'(G)$ is the minimum number of colors needed in a proper edge-coloring of G.*

5.6 Vizing's Theorem

Definition 5.8. *Vizing's Theorem: For any simple graph G, $\chi'(G)$ is either $\Delta(G)$ or $\Delta(G)+1$, where $\Delta(G)$ is the maximum degree of G.*

5.7 Further results on edge-colourings

Definition 5.9. *König's Theorem (bipartite case): In a bipartite graph, $\chi'(G) = \Delta(G)$. (Contextual statement.)*

6 Covers and Matchings

6.1 Covers

Definition 6.1. *A vertex cover of a graph G is a set of vertices S such that every edge of G has at least one endpoint in S. The vertex cover number $\tau(G)$ is the minimum size of a vertex cover. An edge cover is a set of edges that covers all vertices. The edge cover number $\tau'(G)$ is the minimum size of an edge cover.*

6.2 Matchings

Definition 6.2. *A matching in a graph G is a set of edges with no shared endpoints. A maximum matching is a matching of largest possible size. The matching number $\nu(G)$ is the size of a maximum matching. A perfect matching is a matching that covers all vertices of G.*

6.3 Tutte's Theorem

Definition 6.3. *Tutte's Theorem: A graph G has a perfect matching if and only if for every subset $S \subseteq V(G)$, the number of odd components of $G - S$ is at most $|S|$. (Stated for context.)*

7 Algebraic Graph Theory

7.1 Automorphisms

Definition 7.1. *An automorphism of a graph G is an isomorphism from G to itself. The set of all automorphisms of G, denoted $\mathrm{Aut}(G)$, together with the operation of composition, forms a group.*

7.2 Group actions and the Orbit-Stabilizer Theorem

Definition 7.2. *A group action of a group A on a set X is a function $A \times X \to X$ satisfying $e \cdot x = x$ and $(ab) \cdot x = a \cdot (b \cdot x)$ for all $a, b \in A$ and $x \in X$, where e is the identity in A.*

Definition 7.3. *The Orbit-Stabilizer Theorem: For a group action of A on X, the orbit of $x \in X$ under A is $A \cdot x = \{a \cdot x : a \in A\}$, and the stabilizer of x is $\mathrm{Stab}(x) = \{a \in A : a \cdot x = x\}$. Then $|A \cdot x| \cdot |\mathrm{Stab}(x)| = |A|$.*

7.3 The Cauchy-Frobenius-Burnside Theorem

Definition 7.4. *Burnside's Lemma: Let a finite group A act on a set X. Then the number of orbits of X under A is $\frac{1}{|A|} \sum_{a \in A} |\{x \in X : a \cdot x = x\}|$.*